Introduction to LaTeX

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1 Introduction

LaTeXis a language and a tool for creating professionally typeset documents. Writing plain text is as simple as with any other program. **Text** can be *formatted* with *commands*.

```
\textbf{Bolded text.}
\textit{Italisized text.}
\underline{Underlined text.}
\emph{Emphasized text}
```

2 Document Structure

In LaTeXthe document can be divided into sections with the following commands. This document utilizes only sections and subsections.

Command	Level	Comment
part	-1	not in letters
chapter	0	only books and letters
section	1	not in letters
subsection	2	not in letters
subsubsection	3	not in letters
paragraph	4	not in letters
subparagraph	5	not in letters

3 Lists

There are two types of lists: ordered and unordered.

3.1 Ordered Lists

Lesson structure

1. Introduction
2. Syntax
3. Challenge

```
Lesson structure
\begin{enumerate}
        \item Introduction
        \item Syntax
        \item Challenge
\end{enumerate}
```

3.2 Unordered Lists

IB subjects

- Finnish
- English
- Mathematics
- Economics
- History
- Chemistry
- Physics
- Biology
- Computer Science

```
IB subjects
\begin{itemize}
        \item Finnish
        \item English
        \item Mathematics
        \item Economics
        \item History
        \item Chemistry
        \item Physics
        \item Biology
        \item Computer Science
\end{itemize}
```

4 Math

In LaTeXthere are essentially two ways of writing mathematics. **Inline math** is a part of text and **display math** can be used for more complex equations requiring the whole width of the screen. The Pythagorean theorem ($a^2+b^2=c^2$) is one of the most well known pieces of mathematics. Another popular example is the quadratic formula: $s=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$. However, sometimes math requires more space and display math is needed.

$$
\begin{aligned}
& \lim_{ x \to -\infty } \frac{e^{-x}-3e^{-2x}}{e^{-2x}-4} \\
= & \lim_{ x \to -\infty } \frac{e^{-x}-3e^{-2x}}{e^{-2x}-4} \cdot \frac{e^{2x}}{e^{2x}} \\
= & \lim_{ n \to -\infty } \frac{e^{x}-3e^{0}}{e^{0}-4e^{2x}} \\
= & \lim_{ x \to -\infty } \frac{e^{x}-3}{1-4e^{2x}} \\
= & \frac{-3}{1} \\
= & -3
\end{aligned}
$$

```
\begin{align*}
& \lim_{ x \to -\infty } \frac{e^{-x}-3e^{-2x}}{e^{-2x}-4} \\
= & \lim_{ x \to -\infty } \frac{e^{-x}-3e^{-2x}}{e^{-2x}-4} \cdot \frac{e^{2x}}{e^{2x}} \\
= & \lim_{ n \to -\infty } \frac{e^{x}-3e^{0}}{e^{0}-4e^{2x}} \\
= & \lim_{ x \to -\infty } \frac{e^{x}-3}{1-4e^{2x}} \\
= & \frac{-3}{1} \\
= & -3
\end{align*}
```

5 Graphs

The **TikZ** package can be used to create graphics.

5.1 Setup

```
\usepackage{pgfplots}

%General style%
\pgfplotsset{every axis/.append style={
        axis lines = left,
        xmin = 0,
        xmax = 10,
        ymin = 0,
        ymax = 10,
        xlabel style = {anchor = west},
        ylabel style = {anchor  =south},
    clip = false,
    ticks = none,
}}

%Functions%
\pgfplotsset{every axis plot/.append style={
        color=black,
        samples=100,
        thick
}}

%Usage%
\begin{document}
...
\begin{tikzpicture}
  \begin{axis}
        ...
  \end{axis}
\end{tikzpicture}
...
\end{document}
```

5.2 Lines

```
\draw [options] (0,0) -- (10,10);
```

alternatively

```
\addplot [options] coordinates {(0,0) (10,10)};
```

5.3 Arrows

```
\draw [->] (0,0) -- (10,10);
\draw [<-] (0,0) -- (10,10);
\draw [<->] (0,0) -- (10,10);
```

5.4 Polygons

```
\fill[options] (0,0) -- (0,10) -- (10,10) -- (10,0) -- (0,0);
```

5.5 Labels

The location of labels can be specified with coordinates or as relative to some other shape. Note: LaTeXcan be used in labels.

5.5.1 Isolated Labels

```
\node [left]  at (0,0) {Label};
\node [right] at (0,0) {Label};
\node [above] at (0,0) {Label};
\node [below] at (0,0) {Label};

\node at (0,0) {Label};
```

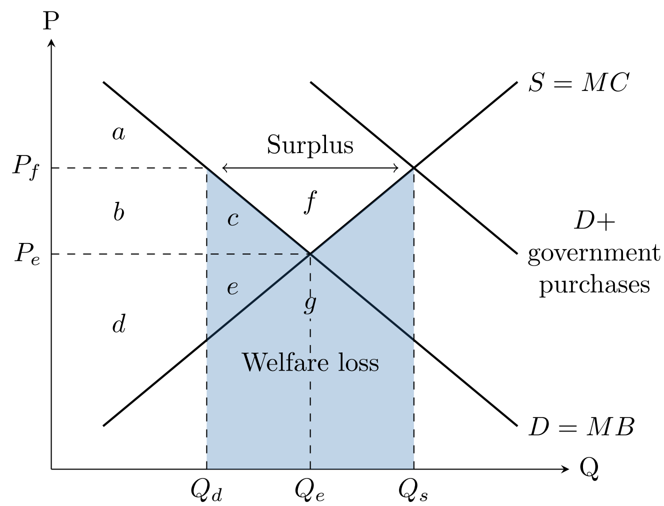
5.5.2 Relative Labels

```
\addplot[] coordinates{(0,0) (10,10)} node[right]{$f$};
```

5.6 Functions

```
\addplot[] {x^2};
```

5.7 Example



P
$S = MC$
a
Surplus
P_f
f
b
c
$D+$
government
purchases
P_e
e
g
d
Welfare loss
$D = MB$
Q
Q_d
Q_e
Q_s

```
\begin{tikzpicture}
  \begin{axis}
    \node [right] at (current axis.right of origin) {Q};
    \node [above] at (current axis.above origin) {P};

    \addplot[] coordinates{(1,9) (9,1)} node[right]{$D=MB$};
    \addplot[] coordinates{(1,1) (9,9)} node[right]{$S=MC$};
    \addplot[] coordinates{(5,9) (9,5)} node[right, align = center]{$D+$ \\ government \\ pu

    \node [left] at (0,5) {$P_e$};
    \node [left] at (0,7) {$P_f$};
    \node [below] at (3,0) {$Q_d$};
    \node [below] at (5,0) {$Q_e$};
    \node [below] at (7,0) {$Q_s$};

    \fill[blue, opacity = 0.3] (3,7) -- (5,5) -- (7,7) -- (7,0) -- (3,0) -- (3,7);

    \addplot[dashed, thin] coordinates {(0,5) (5,5)};
    \addplot[dashed, thin] coordinates {(5,0) (5,2.2)};
    \addplot[dashed, thin] coordinates {(5,2.7) (5,3.5)};
    \addplot[dashed, thin] coordinates {(5,4) (5,5)};

    \addplot[dashed, thin] coordinates {(0,7) (3,7)};
    \addplot[dashed, thin] coordinates {(3,0) (3,7)};
    \addplot[dashed, thin] coordinates {(7,0) (7,7)};

    \node at (1.3,7.8) {$a$};
    \node at (1.3,6) {$b$};
    \node at (3.5,5.8) {$c$};
    \node at (1.3,3.4) {$d$};
    \node at (3.5,4.2) {$e$};
    \node at (5,6.2) {$f$};
    \node at (5,3.8) {$g$};

    \draw [<->] (3.3,7) -- (6.7,7);
    \node at (5,7.5) {Surplus};

    \node at (5,2.5) {Welfare loss};
  \end{axis}
\end{tikzpicture}
```

6 Challenge